

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{4}{81} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_d^{\mathbf{P}}] \delta_{i1i2} - \frac{5}{54} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_d^{\mathbf{P}}] \delta_{i1i2} + \frac{16}{405} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_d^{\mathbf{P}}] \delta_{i1i2} + \right. \\
& \frac{4}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_e^{\mathbf{P}}] \delta_{i1i2} - \frac{5}{18} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_e^{\mathbf{P}}] \delta_{i1i2} + \frac{16}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_e^{\mathbf{P}}] \delta_{i1i2} + \\
& \frac{2}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_l^{\mathbf{P}}] \delta_{i1i2} - \frac{5}{36} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_l^{\mathbf{P}}] \delta_{i1i2} + \frac{8}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_l^{\mathbf{P}}] \delta_{i1i2} + \\
& \frac{2}{81} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_q^{\mathbf{P}}] \delta_{i1i2} - \frac{5}{108} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_q^{\mathbf{P}}] \delta_{i1i2} + \frac{8}{405} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_q^{\mathbf{P}}] \delta_{i1i2} + \\
& \frac{16}{81} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_u^{\mathbf{P}}] \delta_{i1i2} - \frac{10}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_u^{\mathbf{P}}] \delta_{i1i2} + \frac{64}{405} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_u^{\mathbf{P}}] \delta_{i1i2} + \\
& \frac{2}{27} g_1^4 \text{LF}_{3,0} [\tilde{\mu}] \delta_{i1i2} + \frac{1}{9} g_1^4 \text{LF}_{4,-1} [\tilde{\mu}] \delta_{i1i2} - \frac{16}{135} g_1^4 \text{LF}_{5,-2} [\tilde{\mu}] \delta_{i1i2} + \\
& \frac{2}{81} g_1^4 \text{LF}_{2,1,0} [m_1, m_u^{i1}] \delta_{i1i2} + \frac{2}{81} g_1^4 \text{LF}_{2,2,-1} [m_1, m_u^{i1}] \delta_{i1i2} - \frac{4}{81} g_1^4 \text{LF}_{3,1,-1} [m_1, m_u^{i1}] \delta_{i1i2} + \\
& \frac{2}{81} g_1^4 \text{LF}_{4,1,-2} [m_1, m_u^{i1}] \delta_{i1i2} + \frac{2}{81} g_1^4 \text{LF}_{2,1,0} [m_1, m_u^{i2}] \delta_{i1i2} + \frac{2}{81} g_1^4 \text{LF}_{2,2,-1} [m_1, m_u^{i2}] \delta_{i1i2} + \\
& \frac{4}{81} g_1^4 \text{LF}_{3,1,-1} [m_1, m_u^{i2}] \delta_{i1i2} + \frac{2}{81} g_1^4 \text{LF}_{4,1,-2} [m_1, m_u^{i2}] \delta_{i1i2} + \\
& \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_3, m_u^{i1}] \delta_{i1i2} + \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,2,-1} [m_3, m_u^{i1}] \delta_{i1i2} - \\
& \frac{4}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_3, m_u^{i1}] \delta_{i1i2} + \frac{2}{27} g_1^2 g_3^2 \text{LF}_{4,1,-2} [m_3, m_u^{i1}] \delta_{i1i2} + \\
& \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_3, m_u^{i2}] \delta_{i1i2} + \frac{2}{27} g_1^2 g_3^2 \text{LF}_{2,2,-1} [m_3, m_u^{i2}] \delta_{i1i2} - \\
& \frac{4}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_3, m_u^{i2}] \delta_{i1i2} + \frac{2}{27} g_1^2 g_3^2 \text{LF}_{4,1,-2} [m_3, m_u^{i2}] \delta_{i1i2} + \\
& \frac{2}{9} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{2,2,0} [m_d^r, m_q^{\mathbf{P}}] \delta_{i1i2} - \frac{1}{9} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{3,2,-1} [m_d^r, m_q^{\mathbf{P}}] \delta_{i1i2} + \\
& \frac{2}{9} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{2,2,0} [m_e^r, m_l^{\mathbf{P}}] \delta_{i1i2} - \frac{1}{9} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{3,2,-1} [m_e^r, m_l^{\mathbf{P}}] \delta_{i1i2} - \\
& \frac{2}{9} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{3,1,0} [m_l^{\mathbf{P}}, m_e^r] \delta_{i1i2} - \frac{2}{9} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{3,2,-1} [m_l^{\mathbf{P}}, m_e^r] \delta_{i1i2} + \\
& \frac{1}{2} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{4,1,-1} [m_l^{\mathbf{P}}, m_e^r] \delta_{i1i2} - \frac{2}{9} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{5,1,-2} [m_l^{\mathbf{P}}, m_e^r] \delta_{i1i2} - \\
& \frac{2}{9} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{3,1,0} [m_q^{\mathbf{P}}, m_d^r] \delta_{i1i2} - \frac{2}{9} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\mathbf{P}}, m_d^r] \delta_{i1i2} + \\
& \frac{5}{6} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{4,1,-1} [m_q^{\mathbf{P}}, m_d^r] \delta_{i1i2} - \frac{2}{3} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{5,1,-2} [m_q^{\mathbf{P}}, m_d^r] \delta_{i1i2} - \\
& \frac{1}{9} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{2,2,0} [m_q^{\mathbf{P}}, m_u^r] \delta_{i1i2} + \frac{1}{18} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\mathbf{P}}, m_u^r] \delta_{i1i2} - \\
& \frac{1}{18} g_1^2 \overline{y}_u^{\text{pi1}} y_u^{\text{pi2}} \text{LF}_{2,1,0} [m_q^{\mathbf{P}}, \tilde{\mu}] + \frac{1}{36} g_1^2 \overline{y}_u^{\text{pi1}} y_u^{\text{pi2}} \text{LF}_{2,2,-1} [m_q^{\mathbf{P}}, \tilde{\mu}] + \\
& \frac{1}{36} g_1^2 \overline{y}_u^{\text{pi1}} y_u^{\text{pi2}} \text{LF}_{3,1,-1} [m_q^{\mathbf{P}}, \tilde{\mu}] + \frac{1}{9} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{3,1,0} [m_u^r, m_q^{\mathbf{P}}] \delta_{i1i2} + \\
& \frac{1}{9} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{3,2,-1} [m_u^r, m_q^{\mathbf{P}}] \delta_{i1i2} + \frac{1}{3} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{4,1,-1} [m_u^r, m_q^{\mathbf{P}}] \delta_{i1i2} - \\
& \frac{2}{3} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{5,1,-2} [m_u^r, m_q^{\mathbf{P}}] \delta_{i1i2} - \frac{4}{81} g_1^4 \text{LF}_{2,1,0} [m_u^{i1}, m_1] \delta_{i1i2} + \\
& \frac{2}{81} g_1^4 \text{LF}_{3,1,-1} [m_u^{i1}, m_1] \delta_{i1i2} - \frac{4}{27} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_u^{i1}, m_3] \delta_{i1i2} + \\
& \frac{2}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_u^{i1}, m_3] \delta_{i1i2} - \frac{4}{81} g_1^4 \text{LF}_{2,1,0} [m_u^{i2}, m_1] \delta_{i1i2} + \\
& \frac{2}{81} g_1^4 \text{LF}_{3,1,-1} [m_u^{i2}, m_1] \delta_{i1i2} - \frac{4}{27} g_1^2 g_3^2 \text{LF}_{2,1,0} [m_u^{i2}, m_3] \delta_{i1i2} + \\
& \frac{2}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1} [m_u^{i2}, m_3] \delta_{i1i2} - \frac{5}{18} g_1^4 \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] \delta_{i1i2} + \\
& \frac{2}{9} g_1^4 \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] \delta_{i1i2} - \frac{5}{6} g_1^2 g_2^2 \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] \delta_{i1i2} + \frac{2}{3} g_1^2 g_2^2 \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] \delta_{i1i2} + \\
& \frac{1}{36} g_1^2 \overline{y}_u^{\text{pi1}} y_u^{\text{pi2}} \text{LF}_{2,1,0} [\tilde{\mu}, m_q^{\mathbf{P}}] - \frac{11}{36} g_1^2 \overline{y}_u^{\text{pi1}} y_u^{\text{pi2}} \text{LF}_{3,1,-1} [\tilde{\mu}, m_q^{\mathbf{P}}] + \\
& \frac{1}{9} g_1^2 \overline{y}_u^{\text{pi1}} y_u^{\text{pi2}} \text{LF}_{4,1,-2} [\tilde{\mu}, m_q^{\mathbf{P}}] - \frac{1}{18} g_1^4 \text{LF}_{2,1,1,-1} [m_1, m_u^{i1}, \tilde{\mu}] \delta_{i1i2} - \\
& \frac{1}{9} g_1^4 m_1^2 \text{LF}_{2,1,1,0} [m_1, m_u^{i$$